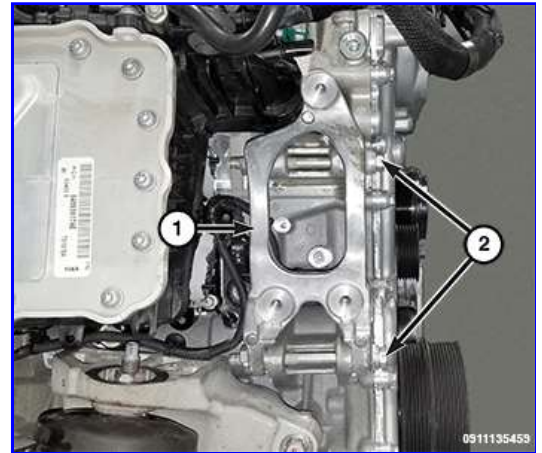


EC1,EC3/ Removal And Installa...

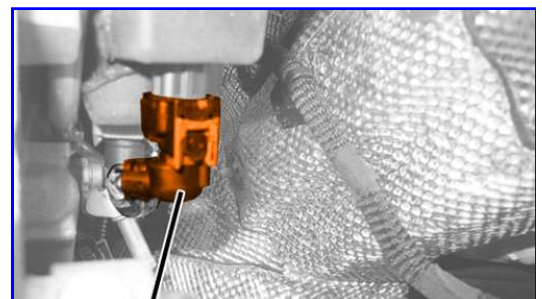
REMOVAL AND INSTALLATION

REMOVAL

1. Raise and support the vehicle ([Refer to 04 - Vehicle Quick Reference/Hoisting/Standard Procedure](#)).
2. Disconnect and isolate the negative battery cable(s)
([Refer to 08 - Electrical/Battery System/Standard Procedure](#)).
3. Drain the engine coolant system ([Refer to 07 - Cooling/Standard Procedure](#)).
4. Drain the turbocharger coolant system ([Refer to 07 - Cooling/Turbocharger/Standard Procedure](#)).
5. Recover the A/C refrigerant system
([Refer to 24 - Heating and Air Conditioning/Plumbing/Standard Procedure](#)).
6. Remove the left front wheel and wheelhouse inner splash shield
([Refer to 23 - Body/Exterior/SHIELD, Splash/Removal and Installation](#)).
7. Remove the cylinder head cover and timing cover
([Refer to 09 - Engine/Valve Timing/COVER\(S\) , Engine Timing/Removal and Installation](#)).
8. If equipped with eTorque, remove the Motor Generator Unit
([Refer to 08 - Electrical/eTorque System/MOTOR, Generator Unit \(MGU\) /Removal and Installation](#)).
9. If equipped with Engine Stop/Start (ESS), remove the generator
([Refer to 08 - Electrical/Charging/GENERATOR/Removal and Installation](#)).
10. If equipped with ESS, remove the two bolts (2) and the generator mounting bracket (1).



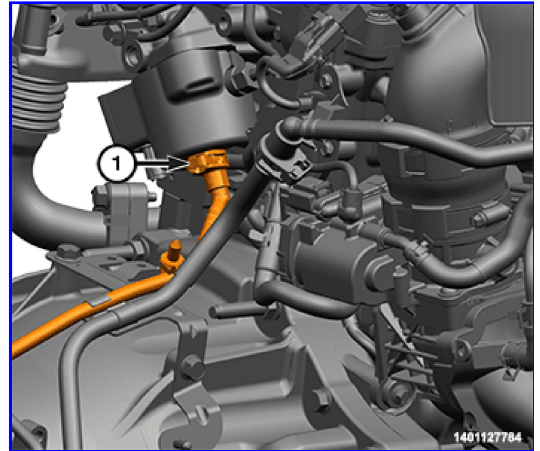
11. Remove the A/C compressor
([Refer to 24 - Heating and Air Conditioning/Plumbing/COMPRESSOR, A/C/Removal and Installation](#)).
12. Remove the rear A/C compressor mounting stud.
13. Remove the catalytic converter
([Refer to 11 - Exhaust System/CONVERTER, Catalytic/Removal and Installation](#)).





14. Disconnect the fuel pressure regulator electrical connection (1).

15. Disconnect the front fuel supply pipe quick-coupling (1) from the fuel injection pump.

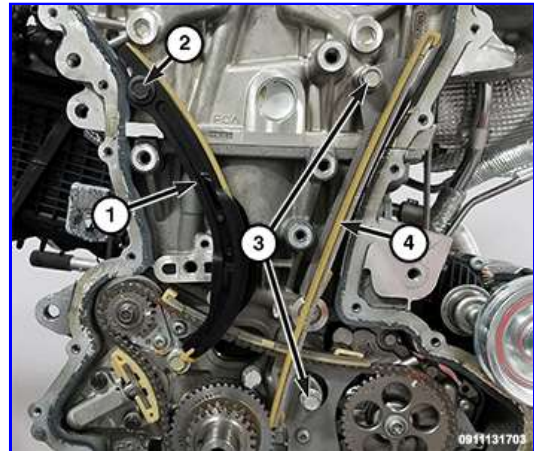


16. Remove the timing chain

(Refer to 09 - Engine/Valve Timing/CHAIN and SPROCKETS, Timing/Removal and Installation) .

17. Remove the bolt and bushing (2) and remove the cam chain tensioner arm (1).

18. Remove the bolts (3) and remove the cam chain guide (4).



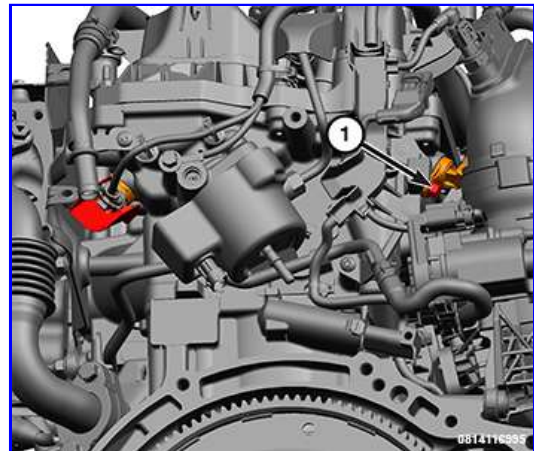
19. Remove the camshafts with the variable valve timing phasers attached

(Refer to 09 - Engine/Cylinder Head/CAMSHAFT, Engine/Removal and Installation) .

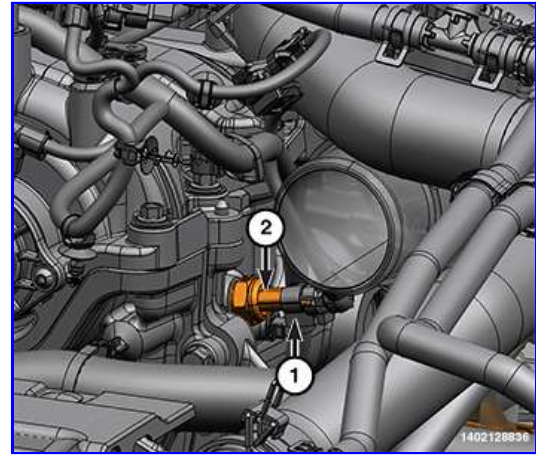
20. Remove the rocker arms and hydraulic lifters

(Refer to 09 - Engine/Cylinder Head/LIFTER(S) , Hydraulic/Removal and Installation) .

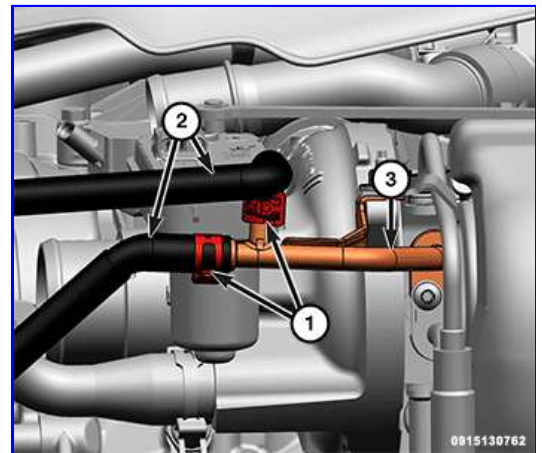
21. Disconnect the intake and exhaust camshaft position sensor electrical connections (1).



22. Disconnect the EGR temperature sensor wire harness connector (1).



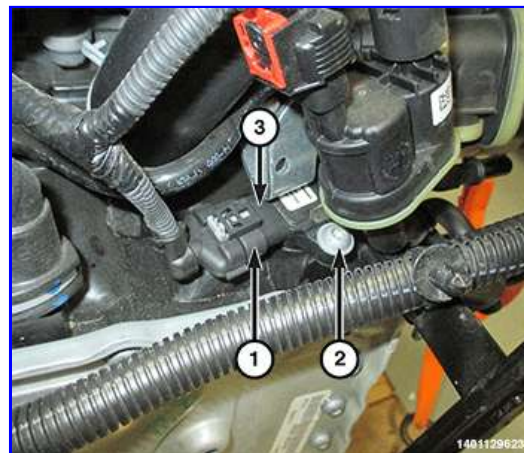
23. Remove the turbocharger oil supply and return pipes (Refer to 09 - Engine/Turbocharger System/LINES and HOSES, Turbocharger, Oil and Coolant/Removal and Installation).
24. Loosen the band clamps (1) and disconnect the hoses (2) from the turbocharger coolant return pipe (3).



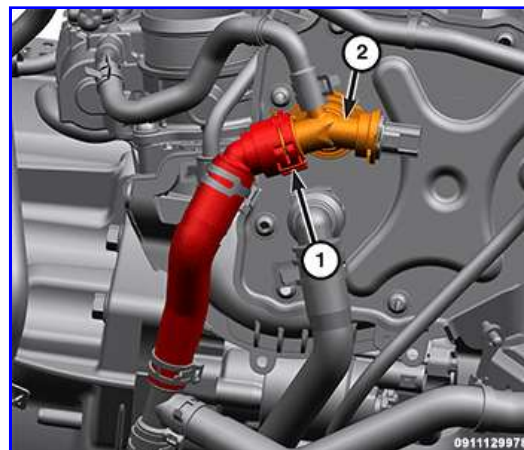
25. Disconnect the electrical connection (1) from the charge air cooler coolant temperature sensor.
26. Disengage seven wire harness retainers (2) and reposition the wire harness.



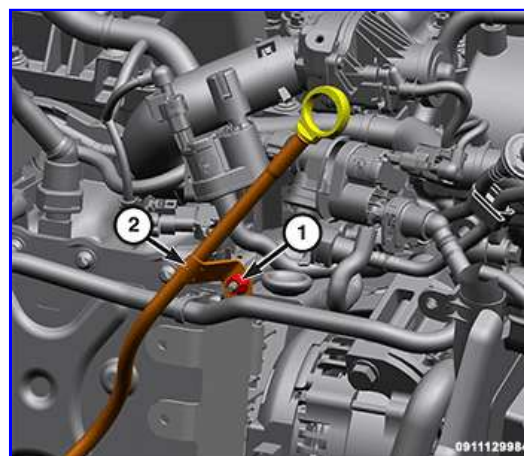
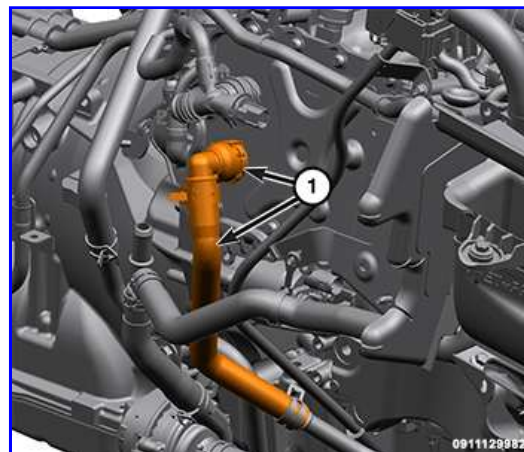
27. Disconnect the Temperature and Manifold Air Pressure (TMAP) wire harness connection (1).



28. Disengage the quick-connect fitting, remove the coolant outlet hose (1) from the Y-fitting (2) and position aside.



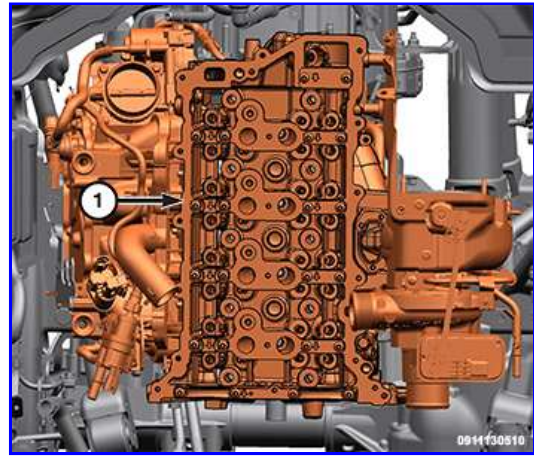
29. Disconnect the quick connector (1) and remove the coolant return pipe from the charge air cooler.



30. Remove nut (1) securing oil dipstick tube (2) and position it aside.

31. Remove the side bolt securing the cylinder head to the crankcase.

32. Remove the inner bolts and remove the cylinder head assembly (1) with the intake manifold and turbocharger attached.

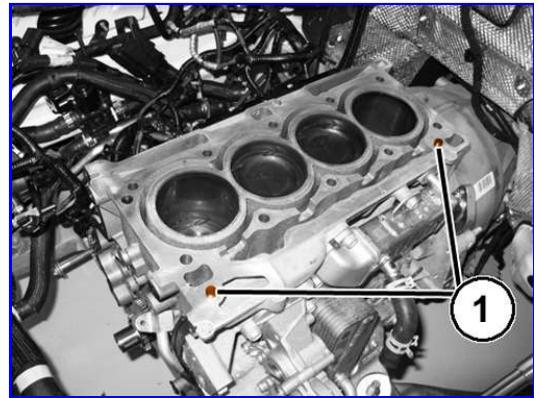


33. Remove the head gasket from the locating pins.

34. Using suitable pliers, remove the oil supply filter (1).



35. Check the locating pins (1).



36. If required, remove the following components from the cylinder head:

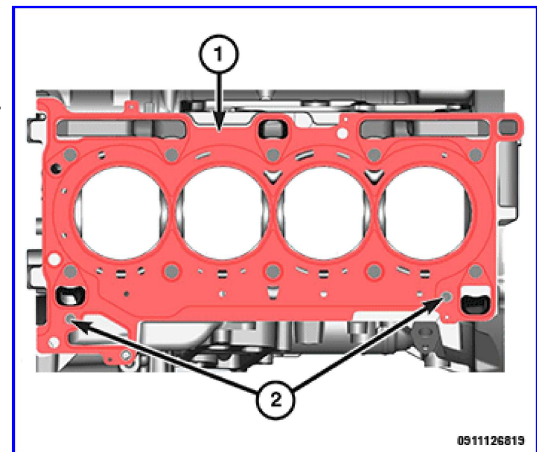
- Intake and exhaust camshaft position sensors
(Refer to 08 - Electrical/Ignition Control/SENSOR, Camshaft Position/Removal and Installation)
- EGR temperature sensor
(Refer to 25 - Emissions Control/Exhaust Gas Recirculation/SENSOR, EGR Temperature/Removal and Installation)
- Fuel injection pump
(Refer to 14 - Fuel System/Fuel Delivery/PUMP, Fuel Injection/Removal and Installation)
- Intake manifold (Refer to 09 - Engine/Manifolds/MANIFOLD, Intake/Removal and Installation)
- Turbocharger (Refer to 09 - Engine/Turbocharger System/TURBOCHARGER/Removal and Installation)

INSTALLATION

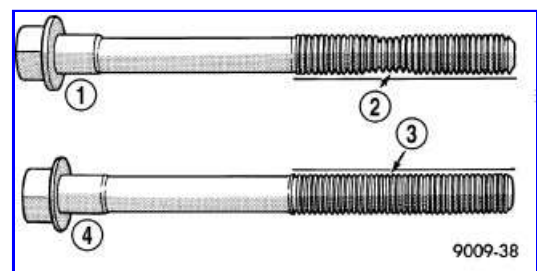
1. If removed, install the fuel injection pump to the cylinder head
(Refer to 14 - Fuel System/Fuel Delivery/PUMP, Fuel Injection/Removal and Installation).
2. Clean and prepare all engine gasket surfaces (Refer to 09 - Engine/Standard Procedure).
3. Install the oil supply filter (1) in the cylinder head.



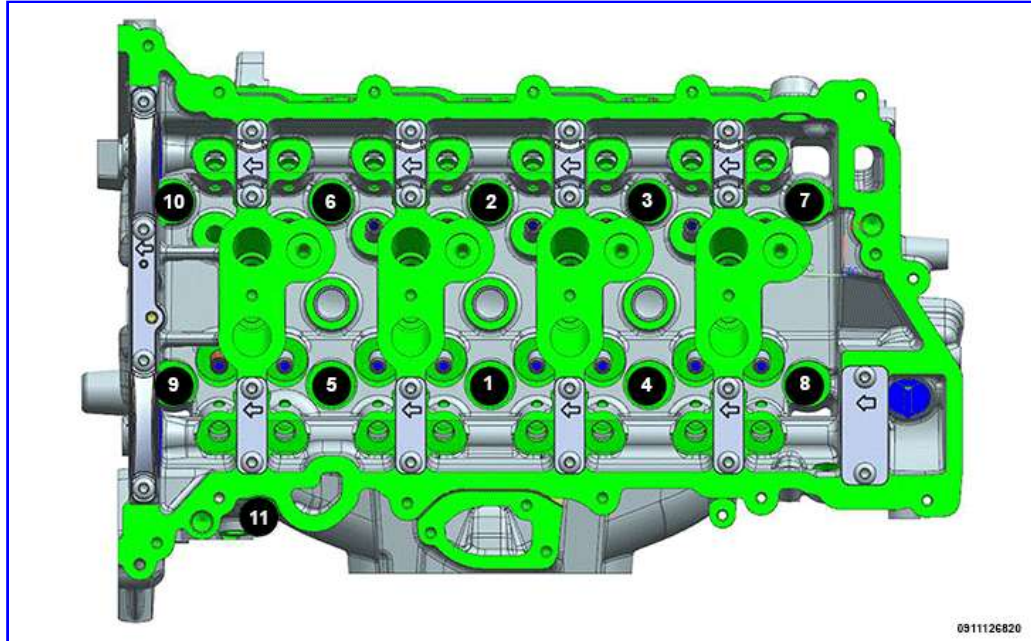
4. Install a new cylinder head gasket (1) onto the locating pins (2).
5. Place the complete cylinder head in position onto the locating pins (2).

**CAUTION:**

The cylinder head bolts are tightened using a torque plus angle procedure. The bolts must be examined BEFORE reuse. If the threads are necked down the bolts must be replaced.



6. Check the cylinder head bolts for necking by holding a scale or straight edge against the threads. If all the threads do not contact the scale (2) the bolt must be replaced.



7. Install ten inner head bolts (1 - 10) hand tight.

8. Install one outer head bolt (11) hand tight.

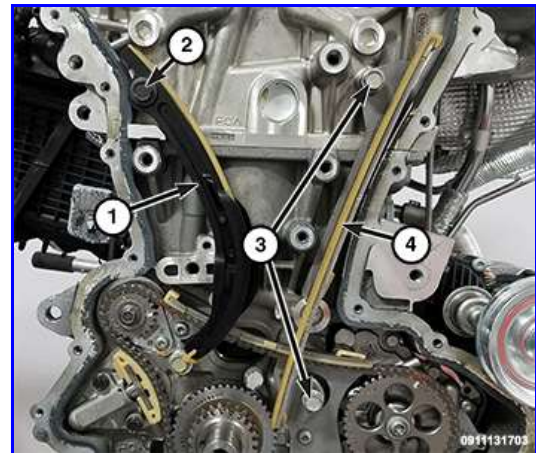
9. Tighten the ten inner cylinder head bolts (1 - 10) in sequence, following this 4 step torque plus angle method. Tighten according to the following torque values:

- Step 1: All to 20 N·m (15 ft. lbs.)
- Step 2: All to 55 N·m (40 ft. lbs.)
- Step 3: All again to 55 N·m (41 ft. lbs.)
- Step 4: All an additional 180° Turn **Do not use a torque wrench for this step.**

10. Tighten the one outer cylinder head bolt (11) to 28 N·m (21 ft. lbs.).

11. Install the cam chain guide (4) and tighten the bolts (3) to the proper [\(Refer to 09 - Torque Specifications\)](#).

12. Install the cam chain tensioner arm (1) with the bushing and bolt (2) tighten to the proper [\(Refer to 09 - Torque Specifications\)](#).



13. Install the hydraulic lifters and rocker arms

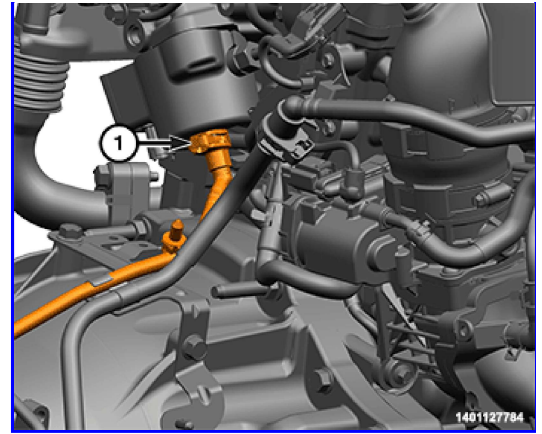
[\(Refer to 09 - Engine/Cylinder Head/LIFTER\(S\) , Hydraulic/Removal and Installation\)](#).

14. Install the camshafts with the variable valve timing phasers still attached

[\(Refer to 09 - Engine/Cylinder Head/CAMSHAFT, Engine/Removal and Installation\)](#).

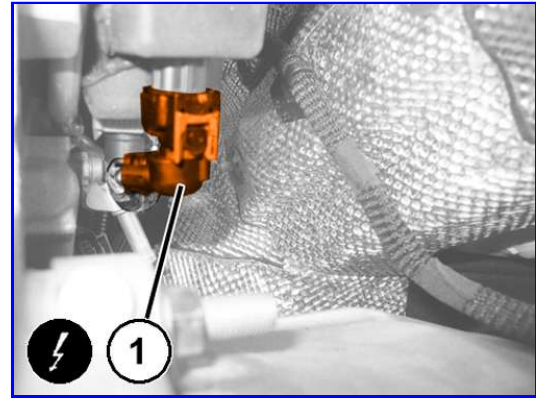
15. Install the cam chain

[\(Refer to 09 - Engine/Valve Timing/CHAIN and SPROCKETS, Timing/Removal and Installation\)](#).



16. Connect the front fuel supply pipe quick coupling (1) to the high pressure pump.

17. Connect the electrical connection (1) for the fuel pressure regulator.



18. If removed, install the following components to the cylinder head:

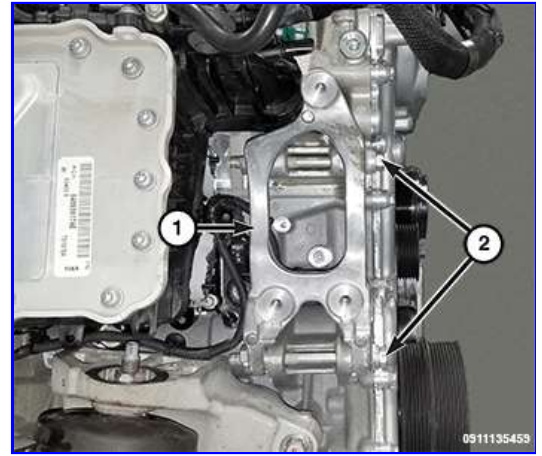
- Intake and exhaust camshaft position sensors
(Refer to 08 - Electrical/Ignition Control/SENSOR, Camshaft Position/Removal and Installation) .
- EGR temperature sensor
(Refer to 25 - Emissions Control/Exhaust Gas Recirculation/SENSOR, EGR Temperature/Removal and Installation)
- Intake manifold (Refer to 09 - Engine/Manifolds/MANIFOLD, Intake/Removal and Installation) .
- Turbocharger (Refer to 09 - Engine/Turbocharger System/TURBOCHARGER/Removal and Installation) .

19. Install the catalytic converter
(Refer to 11 - Exhaust System/CONVERTER, Catalytic/Removal and Installation) .

20. Install the rear A/C compressor mounting stud and tighten to the proper
(Refer to 24 - Torque Specifications) .

21. Install the A/C compressor
(Refer to 24 - Heating and Air Conditioning/Plumbing/COMPRESSOR, A/C/Removal and Installation) .

22. If equipped with ESS, install the generator mounting bracket (1) and tightened the two bolts (2) to the proper
(Refer to 09 - Torque Specifications) .



23. If equipped with ESS, install the generator
(Refer to 08 - Electrical/Charging/GENERATOR/Removal and Installation).
24. If equipped with eTorque, install the Motor Generator Unit
(Refer to 08 - Electrical/eTorque System/MOTOR, Generator Unit (MGU) /Removal and Installation).
25. Install the timing cover, cylinder head cover, accessory drive belt and vacuum pump
(Refer to 09 - Engine/Valve Timing/COVER(S) , Engine Timing/Removal and Installation).
26. Install the left front wheelhouse inner splash shield and wheel
(Refer to 23 - Body/Exterior/SHIELD, Splash/Removal and Installation).
27. Evacuate and charge the A/C refrigerant system
(Refer to 24 - Heating and Air Conditioning/Plumbing/Standard Procedure).
28. Fill the turbocharger coolant system (Refer to 07 - Cooling/Turbocharger/Standard Procedure).
29. Fill the engine coolant system (Refer to 07 - Cooling/Standard Procedure).
30. If removed, install the oil filter and fill the engine crankcase with the proper oil to the correct level (Refer to 04 - Vehicle Quick Reference/Capacities and Recommended Fluids/Specifications).
31. Connect the negative battery cable(s) (Refer to 08 - Electrical/Battery System/Standard Procedure).
32. Remove the support and lower the vehicle.

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